DG-57
Service Manual

PIPETTING SYSTEM VERIFICATION PROCEDURE

Procedure description
This procedure is to be used by user to test that the Pipetting system maintains adequate capability to handling samples, diluents and reagents as required The procedure consists on running a test on a set of 10 cards that dispenses 50 µl of saline solution on one well of each card. Cards are weighed individually before and after dispensing, and differences are used to evaluate accuracy and precision. Test file *.GRU* is available in Annex folder. Note: Document REGD-0000833_v6 replaces document REGD-0000833_v5.

EQUIPMENT

1. 10 test tubes of 15 x 100 mm filled with saline solution (0.9%).
2. 10 cards (Neutral type).
3. Precision laboratory scale, resolution 0.1 µg.
4. Get *S050DISP_8DG.GRU* (DG Gel card) file.

SET UP

1. Turn instrument on.
2. Execute WADiana C. Software.
3. Enable test *S050DISP_8DG.GRU* from Configuration Tab.
4. Wait until the instrument is initialized properly.

PROCEDURE

Process	Expected Results
1. Get and label 10 test tubes and place them in samples carrousel (position 1 to 10).	- - -
2. Get a precision microgram analytical balance. Adjust the weight to zero prior to take measurements.	- - -
3. Get 10 cards and place them individually on weighing dish. Record the number of each card in the column (1) and their weight in the column (2) of the table results (see annexed table).	Column (1) and (2) are filled.
4. Place all cards inside the incubator block, positions 1 to 10.	- - -
5. Execute *S050DISP_8DG.GRU.* Wait until test is completed.	The test is performed.
6. Get all cards (1 to 10) from the incubator. Place individually each card on weighing dish and record the value obtained in column (3).	Column (3) is filled.
7. To determine the weight of the water in the card after each dispensing, subtract from the card's weight of column (3) the card's weight of column (2) and write the result in column (4).	Column (4) is filled.
8. Calculate the Mean and the Standard Deviation of the column (4) values. Write the calculation results on cells (5) and (6) respectively.	Mean value (5) and S.D. (6) results are filled
9. Calculate the *Precision* (CV) as follow and write the result in (7): $$CV = \frac{Standard\ Deviation\ (SD)}{Mean\ Value} x100$$	<=3%
10. Calculate the *Accuracy* as follow and write the result in (8): $$\%\,Accuracy = \frac{MeanVolume - TestVolume}{TestVolume} x100$$	<=±10%

TABLE of RESULTS(*)

Equipment identification:	
Analytical balance id. :	
Test used:	
Volume of the test:	___ μL
Temperature room:	

Readings:

(1) Card Number	(2) Balance tare Reading (microg)	(3) Balance reading after dispensing	(4) Weight dispensing (microg)

Mean pipeting (5) S.D. (6) PRECISION CV% (7) ACCURACY % (8)	

Date:
Done by:

(*) The pipetting accuracy and precision test is based on the gravimetric test, assuming that errors introduced by saline water density and environmental temperature are negligible.

DOCUMENT CHANGE HISTORY

Document change history			
Version	**Date**	**Change**	**Rationale**
1.0	2005/02/18	Document REGD-0000833 replaces document 57060V0M000.	Uploaded document to GxPharma.
2.0	2005/05/09	Eliminated GRU list and added external reference to it, updated test list with DG-Gel.	New DG Gel cards.
3.0	2005/07/28	Updated procedure description and added reference to gravimetric test.	Updated procedure description and added note to make it easier.
4.0	2011/04/28	- Eliminated references to DianaGel and Diamed cards. - Changed *S050DISP_NXX.GRU* (*N* is a number and *XX* are letters dependents from card used, for more details see equipment section) by *S050DISP_8DG.GRU.*	Service manual intended for Grifols market.
4.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
5.0	2011/07/20	Document REGD-0000833_v5 replaces document REGD-0000833_v4.	Document version changed.
5.0	2011/07/20	Document title.	Commercial reference substituted by project reference.
5.0	2011/07/20	Document header.	Increase information to easy document search.
6.0	See document header.	CV value <=± 3% replaced by <=3%	CV value must be <=3%